

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Scenario-Based Questions

Code of Conduct:

I M. Pranay (192525382) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

M. Pranay

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Questions

1. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

- (a) Interpret the scenario and explain the importance of boundary detection.
- (b) Identify key challenges in detecting boundaries.
- (c) Apply a suitable edge detection method.
- (d) Justify your selection with logical reasoning.
- (e) Present your explanation clearly and systematically.

A.

(a) Interpretation and Importance

In medical images, boundary detection is used to identify the edges between different tissues or organs. Accurate boundary detection helps doctors identify abnormalities, measure organs, and plan treatments.

(b) Key Challenges

- Image noise
- Low contrast between tissues
- Blurred boundaries
- Overlapping structures

(c) Suitable Edge Detection Method

Canny Edge Detection

Steps:

1. Apply Gaussian filter to remove noise.
2. Calculate image gradients.
3. Perform Non-Maximum Suppression.
4. Apply Double Threshold.
5. Track edges using Hysteresis.

(d) Justification

- High accuracy
- Removes noise effectively
- Produces thin and continuous edges
- Reduces false edge detection

(e) Conclusion

Canny Edge Detection is the most suitable method because it provides accurate, continuous, and noise-free boundaries required for medical image analysis.

2. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

- (a) Interpret the scenario and explain segmentation requirements.
- (b) Identify key features enabling separation.
- (c) Apply a suitable segmentation technique.
- (d) Justify your decision logically.
- (e) Present your response clearly.

A.

(a) Interpretation

The objects have different intensity values from the background, making them easy to separate.

(b) Key Features

- High intensity difference
- Uniform background

- Clear object boundaries

(c) Suitable Segmentation Technique

Threshold Segmentation (Global Thresholding)

Steps

1. Read the image.
2. Select a threshold value.
3. Pixels greater than the threshold become the object.
4. Remaining pixels become the background.

(d) Justification

- Simple implementation
- Fast execution
- Works well for images with clear intensity differences
- Computationally efficient

(e) Conclusion

Threshold segmentation is ideal because it quickly separates objects from the background when intensity differences are clear.

3. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

- (a) Interpret the scenario and explain segmentation challenges.
- (b) Identify key issues such as intensity overlap.
- (c) Apply an appropriate advanced segmentation method.
- (d) Justify how the method handles complexity.
- (e) Provide a structured and clear explanation.

A.

(a) Interpretation

The image contains several regions with similar or overlapping intensity values, making segmentation difficult.

(b) Key Issues

- Overlapping intensities
- Noise
- Irregular object shapes
- Weak boundaries

(c) Suitable Advanced Segmentation Method

Watershed Segmentation

Steps

1. Remove noise.
2. Compute image gradients.
3. Mark foreground and background.
4. Apply Watershed algorithm.
5. Obtain segmented regions.

(d) Justification

- Separates touching objects
- Handles overlapping intensity values
- Produces accurate region boundaries
- Suitable for complex images

(e) Conclusion

Watershed Segmentation effectively separates complex regions even when intensity values overlap.

4. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

- (a) Interpret the scenario and explain why edges are broken.
- (b) Identify key issues affecting boundary continuity.
- (c) Apply a suitable method to improve edge connectivity.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

A.

(a) Interpretation

After edge detection, boundaries are incomplete because of image noise or weak edge responses.

(b) Key Issues

- Noise
- Weak gradients
- Poor threshold selection
- Missing edge pixels

(c) Suitable Method

Morphological Closing (Dilation followed by Erosion)

Steps

1. Detect edges.
2. Apply Dilation.
3. Apply Erosion.
4. Obtain connected boundaries.

(d) Justification

- Connects broken edge segments
- Fills small gaps
- Preserves object shape
- Improves boundary continuity

(e) Conclusion

Morphological Closing improves edge connectivity and creates continuous object boundaries.

5. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

- (a) Interpret the scenario and define the combined requirement.
- (b) Identify key issues involving noise and edge preservation.
- (c) Apply an appropriate sequence of techniques.
- (d) Justify the order and selection with strong reasoning.
- (e) Present your response in a clear and structured format.

A.

(a) Interpretation

The industrial inspection system must remove image noise while preserving important edges to detect product defects accurately.

(b) Key Issues

- Noise affects detection accuracy.
- Excessive smoothing removes useful edges.
- Edge preservation is essential for identifying defects.

(c) Appropriate Sequence of Techniques

1. **Median Filter** – Remove noise while preserving edges.
2. **Canny Edge Detection** – Detect important edges.
3. **Morphological Operations** – Improve edge continuity if necessary.

(d) Justification

- Median Filter removes salt-and-pepper noise without blurring edges.
- Canny Edge Detection provides accurate and continuous edges.
- Morphological operations connect broken edges, improving defect detection.

(e) Conclusion

The sequence **Median Filter → Canny Edge Detection → Morphological Operations** ensures effective noise removal while preserving critical edges, making it suitable for industrial defect inspection.